Reliable Annotation and Evaluation of Malaysian Malay-English Code-Mixed Sentiment Data: A Critical Review of an Unresolved Problem

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Abstract

Malaysian social media users mix English and Malay inside single messages, producing the contact variety known locally as Manglish. The studies reviewed here usually treat sentiment analysis of Malay-English mixed writing as a modelling problem. This review indicates that the more limiting constraint is earlier in the process, in the quality of the sentiment labels and in the evidence offered that those labels are sound. The review covers annotation practice and reliability evaluation for Malaysian Malay-English code-mixed sentiment data. Comparative benchmarking of sentiment classifiers is outside its scope. Three contributions follow. First, the annotation strategies used for low-resource code-mixed text are organised into three tiers, and the trade-off between label fidelity and scalability is examined across them. Second, two sources of neutral-label skew are separated, one produced by the labelling tools and the other by genuine uncertainty among annotators. The four Malay-English sentiment resources are then examined against what a reliability claim requires, and none supplies it. Third, the class imbalance in the label distribution is shown to be the condition that makes Cohen's kappa more difficult to interpret. The review closes with a research agenda covering minimum reporting standards, agreement measurement under class imbalance, and the untested assumption that annotators of different types make different errors.

Keywords: *sentiment analysis, code-mixed text, Manglish, annotation reliability, inter-annotator agreement*

1. Introduction

Sentiment analysis, also known as opinion mining, is the automated task of deciding whether a piece of text expresses a positive, negative or neutral opinion (Poria et al., 2023). Organisations use it to measure public opinion at scale, in domains that include consumer products and services, healthcare, political elections and financial services (Nazir et al., 2022). However, the tools used for this task assume that their input is written in one standard language. Joshi et al. (2020) highlight that only a small number of the world's languages are represented in current language technologies, with lexicons, pre-trained models and benchmark corpora concentrated in those few languages. Among these three, a pre-trained model is one that has already learned general patterns from a large body of text, and is then adapted to a task such as sentiment analysis (Devlin et al., 2019). Many of these language resources are developed primarily for individual languages, so text in which two languages alternate within a single message does not meet the

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conditions for which they were designed. Doğruöz et al. (2021) call this alternation code-switching, and report that its effect on model performance varies by task. In their survey, current models perform well on code-switched language identification and part-of-speech tagging, but only slightly better than chance on natural language inference. Winata et al. (2023), surveying a wider set of studies, also report that performance on code-switched text is often lower than performance on monolingual text for the same task, though not uniformly so. Code-switching between English and Malay is common in Malaysian social media text (AlAfnan et al., 2025).

Malaysia is a multilingual and multi-ethnic country. Drawing on Ethnologue figures, Coluzzi (2017) records around 140 languages in use in the country, and reports that code-mixing between a speaker's first languages and English or Malay is widespread in everyday communication. AlAfnan et al. (2025) report that English and Malay are among the languages most frequently combined in Malaysian written communication. The same study describes social media users alternating between the two languages within a single message rather than writing in one language at a time. Two local terms are used for this code-mixed writing, and they refer to different combinations of languages. AlAfnan et al. (2025) describe *Manglish* as a hybrid form that combines Malay and English, whereas Romadhona et al. (2022) describe *Bahasa Rojak* as a variety that combines English, Malay and Chinese. This review is concerned with Malay-English mixing. *Manglish* names that combination, while *Bahasa Rojak* names a broader one that also includes Chinese.

Manglish is not always described in the same terms. AlAfnan et al. (2025) note that it is sometimes viewed as having a negative effect on standard English and Malay rather than as a legitimate language variety in its own right. The structure of *Manglish*, however, is systematic rather than random. Treffers-Daller et al. (2022) analyse the classroom speech of two Malaysian teachers of English and find that Malay-English switching there follows describable syntactic patterns. AlAfnan et al. (2025) report that speakers use *Manglish* for identifiable purposes, which include communicating efficiently, marking identity and building rapport.

The same systematicity appears at the level of spelling. Maskat and Rahman (2020) document the Malay negator *tidak* being written as *x*, and *tidak ada* being written as *takde, xde* or *tade*. They sort this variation into five categories and map two of them back to their standard forms, and this categorisation suggests that the variation follows recurring patterns rather than occurring arbitrarily. A dictionary of standard Malay, such as *Kamus Dewan*, does not list these non-standard forms (Dewan Bahasa dan Pustaka, 2005). Tools that rely on such a dictionary may therefore treat them as out-of-vocabulary items, meaning words absent from the reference lexicon (Maskat and Rahman, 2020). Abu Bakar et al. (2020) describe Malay social media text as misspelled, out-of-vocabulary, ill-formed and noisy. This review reads that description as a statement about the reference standard being applied rather than about the care taken by the writer. These writing conventions are consistent within the observed variation, but they differ from the conventions that standard natural language processing tools were designed to handle.

These differences between Malaysian writing conventions and the assumptions built into standard tools raise a question about the source of the difficulty. When a sentiment model performs poorly on Malay-English mixed text, one common

explanation concerns the model architecture (Doğruöz et al., 2021; Winata et al., 2023). Under this view the model processes code-switched input less effectively than monolingual input, so a larger model, or one further trained on Malay-English or other Southeast Asian data, may perform better. The evidence for this explanation is mixed. Khanuja et al. (2020) report that multilingual models fine-tuned on code-switched data achieve the strongest scores among the systems evaluated on their benchmark. Adilazuarda et al. (2022) find that robustness to code-mixing varies with the embedded language, and attribute the variation to bias in the pre-training corpus. In contrast, Winata et al. (2021) find that pre-trained multilingual models do not necessarily produce high-quality representations of code-switched text. Conneau et al. (2020) describe a trade-off between two effects in multilingual models. Adding languages can improve performance on each of them, an effect they call positive transfer. Beyond a point, however, the modelling capacity available to any individual language falls, an effect they call capacity dilution, and the quality of its representations may decline. Doğruöz et al. (2021) identify a further limitation that is not architectural: the multilingual models in common use are trained mainly on monolingual corpora and may receive little code-switched text during pre-training. Model choice therefore accounts for part of the difficulty but not all of it.

Another source of difficulty arises one step earlier, at the point where sentiment labels are created. Supervised sentiment analysis requires a sentiment label for every message before a model can be trained or evaluated on it. The amount of labelled data available therefore sets a bound on what can be trained and on what can be tested. Poria et al. (2023) describe this reliance on annotated data as often limiting, and note that it has renewed interest in supplying models with external knowledge such as sentiment lexicons. A sentiment label is assigned either by a human annotator or by an automatic labelling method, and this step is called annotation. The labels form the supervision signal used to train a model, and the labels in a test set are the standard against which the model's predictions are compared. Incorrect labels can affect both model training and subsequent evaluation. During training, a model may learn the patterns associated with incorrect labels, which can reduce predictive accuracy (Frénay and Verleysen, 2014). During evaluation, incorrect labels in a test set can bias the measurement of model performance. Northcutt et al. (2021) show that label errors can change which of two models is ranked higher on a benchmark. Bowman and Dahl (2021) name annotation reliability as one of the conditions a sound evaluation benchmark must meet. An incorrect label can therefore have an effect at two separate stages, first when the model is trained and again when its performance is measured.

Those two stages apply to supervised learning in general. For Malaysian Malay-English code-mixed sentiment data in particular, the question is whether the model or the sentiment labels are the greater constraint at present. Three findings point to the labels, and each is developed in the sections that follow. First, this review identifies four Malay-English sentiment resources (Kasmuri and Basiron, 2019; Romadhona et al., 2022; Kong et al., 2023; Mohd Shamsuddin et al., 2024). None of them meets all four of the criteria a general-purpose resource would need. Those criteria are (1) at least ten thousand labelled messages; (2) a label distribution that is not dominated by a single sentiment class; (3) text drawn from everyday social media posts rather than from blogs or reviews; and (4) an assessment of annotation

reliability by human annotators. Each of them satisfies some of these criteria but not all of them. Second, the largest of them is labelled 78.23% neutral (Mohd Shamsuddin et al., 2024). That skew towards the neutral label appears to originate in the labelling method rather than in Malaysian discourse. Third, the studies introducing two of the four resources report no inter-annotator agreement coefficient at all (Kong et al., 2023; Mohd Shamsuddin et al., 2024). Where a coefficient is reported, it was measured on a two-way judgement rather than on the three-way positive, negative and neutral labelling for which the resource is used (Kasmuri and Basiron, 2019; Romadhona et al., 2022).

The second and third findings are closely related rather than independent. Reliability is normally assessed by having independent annotators label the same text and computing a chance-corrected agreement coefficient, such as Cohen's kappa (κ) (Cohen, 1960; Artstein and Poesio, 2008). Cohen's κ becomes difficult to interpret when one category dominates the label distribution, because the agreement expected by chance rises with that skew (Feinstein and Cicchetti, 1990; Byrt et al., 1993). Di Eugenio and Glass (2004) call this effect the prevalence problem. The skew towards the neutral label described in the second finding can therefore make Cohen's κ, the statistic named in the third, more difficult to interpret. Neither correction would be sufficient without the other. These four resources are therefore limited both in what they contain and in what is known about the reliability of their labels. That shortage is not confined to them, since Selvaraju et al. (2024) report that comprehensive datasets and annotated resources for code-mixed Malay text are still lacking more generally.

These three findings lead to the question this review examines: why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data have not yet been established. Linguistic description, model architecture and application domain are discussed only where they are relevant to that question. Two related topics are not examined here. The first is comparative benchmarking of sentiment classifiers, and the second is sentiment analysis as a general field. Both have already been surveyed in detail (Poria et al., 2023; Nazir et al., 2022). This review does not propose, implement or evaluate a system, and it does not report the authors' own experimental work. Its contribution is an analysis of why that question remains open, together with a set of directions for future research.

This review makes three contributions:

1. It organises the annotation strategies used for low-resource code-mixed text into three tiers, which are manual, semi-automated and fully automated labelling, and examines the trade-off between label fidelity and scalability across them.
2. It distinguishes two sources of neutral-label skew, one produced by the labelling tools themselves and the other by genuine uncertainty among annotators. It also examines what the studies that introduced the four Malay-English sentiment resources report about how reliably each was annotated.
3. It shows how the linguistic, annotation and evaluation challenges compound one another rather than merely co-occurring, and sets out a research agenda that follows from that interaction.

The literature examined here was identified through searches of the ACL Anthology, IEEE Xplore, Scopus and arXiv, combining terms for the language variety, the sentiment task and annotation quality. Studies that only benchmark classifiers, with no annotation or reliability component, were not included.

The rest of this paper is organised as follows. Section 2 explains why Malaysian code-mixed writing is difficult to annotate. Section 3 examines annotation practice and its failure modes, and Section 4 examines the models that are used as annotators. Section 5 discusses how annotation reliability is evaluated. Section 6 brings these findings together and sets out directions for future research, and Section 7 concludes the review.

2. Why Malaysian Code-Mixed Text Is Difficult to Annotate

Section 1 identified the sentiment labels, rather than the model, as the greater constraint on sentiment analysis of Malaysian Malay-English text. Annotation practice is examined in Section 3. The text being annotated is described first, because three of its linguistic properties affect the annotation task directly. The first property is that switching between Malay and English occurs inside the sentence. The boundary between the two languages therefore does not show where the opinion in a message is located. The second property concerns the affect of a message, meaning the emotional attitude it expresses. That affect can be carried by Malay particles, and these particles appear in no standard sentiment lexicon. The third property is that phonetic spelling gives a single word many written forms. All three properties are described in the sociolinguistic literature on Malay-English contact (Muysken, 2013; Treffers-Daller et al., 2022; AlAfnan et al., 2025). Each property is presented below together with its consequence for the annotator. The annotator may be a human or an automatic labelling method.

2.1 Switching occurs within the sentence and offers little reliable structural cue

Treffers-Daller et al. (2022) analysed 1,044 switched utterances from the classroom speech of two Malaysian teachers of English. In their data, the majority of the switches occurred within the clause rather than between clauses. Insertion accounted for 33.0% of the switches. In insertion, a word or phrase from one language is placed inside a stretch of the other. Congruent lexicalisation accounted for a further 30.7%. In congruent lexicalisation, words from both languages are combined within a single shared word order. Alternation between whole clauses accounted for 24.8%. The authors attribute this distribution to the structure of the two languages rather than to the style of the speakers. English and Malay share the same basic word order. In both languages, the subject comes before the verb and the verb before the object. Malay also has very little inflectional morphology. Inflectional morphology refers to changes in the form of a word that mark grammatical features such as tense. A speaker can therefore move from one language to the other in the middle of a clause with little grammatical adjustment (Treffers-Daller et al., 2022; Muysken, 2013).

These proportions were measured on spoken classroom data. They should therefore not be taken as the proportions in social media text. The explanation given by the authors, however, concerns the structure of the two languages and not the

medium in which they are used. The same type of switching can therefore be expected in writing. Mohd Shamsuddin et al. (2024) give a written example in the post "Does institut kanser offer free medication cheap chemotherapy". Here the Malay noun phrase *institut kanser* is placed inside an English question. The phrase follows Malay word order, in which the noun comes before its modifier.

The consequence for annotation is that the annotator cannot use the boundary between the two languages to locate the part of the message that carries the sentiment. In monolingual annotation, locating that part is a separate step from judging its polarity. Aspect-based sentiment analysis is one task in which the two steps are separate. In that task, a sentiment label is assigned to each entity or attribute mentioned in a text rather than to the text as a whole (Nazir et al., 2022). In the annotation scheme described by Pontiki et al. (2016), the annotator first marks the words that express the opinion as a span. A span is a marked stretch of text. The annotator then assigns a polarity label to that span. In Malay-English mixed text, the annotator has no language boundary to guide the first step. Deciding which span to judge already requires interpretation. Two annotators may therefore assign different labels because they are judging different parts of the message, not because they judge the sentiment differently. Disagreement about which part of a message to judge is therefore separate from disagreement about its polarity. The first of these two disagreements also arises before the second.

2.2 Affective meaning is carried by particles that belong to neither standard language

Locating the part of the message that carries the opinion is the first difficulty. A second difficulty arises once that part has been located. The word that carries the affect may be a Malay particle. That particle may be attached to a clause that is otherwise entirely English. Muysken (2013) terms this construction back-flagging. Treffers-Daller et al. (2022) define back-flagging as the attachment of a discourse marker or a coordinating conjunction from a speaker's first language to a stretch of speech in the second language. In their data, back-flagging accounts for 121 of the 1,044 switched utterances, or 11.6%. The particle involved most frequently is *lah* (Treffers-Daller et al., 2022). The same study reports that *lah* can attach to a clause that is entirely English, entirely Malay or mixed. Two examples that the study reproduces from earlier descriptive work are "Buy this lah" and "you understand lah kan".

These particles contribute to the tone of a sentence rather than to its factual content. AlAfnan et al. (2025) classify particles as a lexical feature of Manglish. In their account, particles add emphasis or tone and signal politeness or informality. They also list particles among the features that distinguish Manglish from standard English and standard Malay. One function of these particles, according to the same study, is to soften an instruction. The example given is "Don't disturb la, he's busy", in which the particle makes the instruction less direct. Treffers-Daller et al. (2022) describe the same effect for *lah*. They report that *lah* can shift an utterance from a highly assertive tone to a polite request or a friendly encouragement. In both accounts, the particle leaves what the sentence states unchanged. What the particle

alters is the attitude the sentence conveys. That attitude is the property a sentiment label is intended to record.

Two limits on this evidence should be noted. Treffers-Daller et al. (2022) supply frequencies. Their corpus, however, is fifteen hours of recorded classroom teaching by each of two university English lecturers. Their frequencies should therefore not be read as frequencies for social media writing. AlAfnan et al. (2025) describe Malaysian usage more broadly. Their data, however, are scripted dialogue from Malaysian television series such as *Kopitiam* and *KL Gangster*. Those data were examined through qualitative content analysis. That study therefore reports what the particles do, not how frequently they occur. Neither source establishes how frequently particles are used in Malaysian social media posts. No such figure was found among the studies reviewed here. What both sources do establish is that the construction recurs and that it carries affect.

The consequence for annotation is that the particle indicates the tone of a message but does not determine its polarity. A reader of "Don't disturb la, he's busy" can see that the particle softens the instruction. A softened instruction can still be read in two ways. Under a negative reading, the writer is irritated. Under a neutral reading, the writer is being polite. Treffers-Daller et al. (2022) note the same difficulty for discourse markers as a whole, whose specific meanings are often difficult to capture even where their pragmatic functions can be evaluated. The particle therefore carries affect without showing which polarity that affect has. Two annotators may read the same particle differently, and both may be reading it competently.

Disagreement of this kind is not necessarily reduced by adding annotators. Pavlick and Kwiatkowski (2019) find that where an item can be read in more than one justifiable way, the disagreement between annotators is reproducible. The same disagreement appears again as further judgements are collected, and the judgements do not converge on a single label. A corpus built by majority vote over such items therefore records the majority reading as though it were the only justifiable one. The variation that the majority vote conceals is systematic rather than random. An agreement coefficient computed over such items therefore reflects genuine differences in interpretation rather than annotator error alone.

The first property examined in this section produces disagreement about which part of a message to judge. The second property produces disagreement about how to judge that part once it has been located. That disagreement arises because more than one reading of the part is justifiable.

2.3 Phonetic spelling multiplies the surface forms of a single word

The first two properties concern where the sentiment in a message is located and how it should be read. The third property concerns whether the words that carry the sentiment can be recognised at all. Maskat and Rahman (2020) document Malay social media users spelling words as they are pronounced rather than as a dictionary lists them. A single word can therefore appear in many written forms. Their corpus is 6,241 Twitter posts collected between 2014 and 2018. They call the language of these posts Social Media Malay Language and sort it into five categories. The

categories are spelling variants, Malay-English mixed sentences, loan words spelled according to Malay pronunciation, slang-based spellings, and vowel-less words.

Each category contains many forms. Among the spelling variants that Maskat and Rahman (2020) record, *macam mana* appears as *mcmana*, *camne*, *camner*, *mcmner* and *cane*. In the vowel-less category, a word is reduced to its consonants or to a single character. For example, *jangan* is written *jgn* and *pada* is written *pd*. The slang category varies by region. The word *besar* is written *besaq* in the north and *beso* on the east coast. The categories can also combine. The same study gives *x suko* as a form that belongs to the slang and vowel-less categories at once.

The question that matters for annotation is whether this variation can be reversed automatically. Maskat and Rahman (2020) tested this question directly. They attempted to normalise two of the five categories, spelling variants and vowel-less words. To normalise a form means to map it automatically to its standard spelling. Two measures of string similarity were used, n-gram similarity and Levenshtein distance. Each measure scores how closely two spellings resemble one another. The benchmark was a set of 1,176 pairs of similar words covering 435 distinct words. A native Malay speaker identified these pairs by hand. Precision here is the proportion of the matches proposed by a method that were correct. On that benchmark, the highest precision reported for any configuration was 0.43. Most of the configurations reported were lower than that (Maskat and Rahman, 2020). The authors conclude that similarity matching can detect the two categories but that a more sophisticated technique is needed. They also state that normalisation of the remaining three categories is left for future research.

Two limits on this evidence should be noted. The first is that only slang from Selangor was handled. The slang category was therefore tested more narrowly than the regional range the same paper describes. The second limit matters more for this review. English words were removed before the experiment, and code-mixed posts were excluded from the evaluation subsets. The reported precision is therefore a measurement on Malay text with the English removed, not on Malay-English mixed text. No equivalent measurement for mixed text was found among the studies reviewed here.

The consequence for annotation is that human and automatic annotators differ in what they can recognise. The difference can be seen in the study just described. A single native speaker identified 1,176 word pairs by hand. That identification was reliable enough for the result to serve as the ground truth. Automatic matching on the same ground truth reached a highest precision of 0.43. A human reader recognises *x* as *tidak*. The automatic procedure largely did not. The two types of annotator therefore make errors on different inputs and at different rates.

This difference changes what the choice of annotator means. If human and automatic annotation differed only in cost and speed, the choice between them would be a matter of budget. Because the two types make errors on different inputs, the choice also determines which errors a corpus will contain. A corpus labelled entirely by one type of annotator therefore carries that type's systematic errors throughout.

The three properties examined in this section therefore create three separate difficulties for annotation. The first property makes the part of the message that carries the opinion difficult to locate. The second property leaves the reading of that part open once it has been located. The third property makes the words themselves difficult for an automatic annotator to recognise. A human annotator recognises the same words readily. Section 3 examines the annotation strategies that have been used in response to these difficulties. It also examines the trade-off that each strategy accepts.

3. Annotation Practice and Its Failure Modes

Section 2 described what makes Malaysian code-mixed text hard to label. This section examines what has been done about it. The strategies reported in the reviewed studies are set out first, arranged by the trade-off each one accepts. The distribution of labels they produce is examined next, and the practice of pooling several annotators is examined last, because it is the remedy most often proposed, and the one the regional evidence tests most directly (Artstein and Poesio, 2008; Uma et al., 2021).

3.1 Annotation strategies and the fidelity-scalability trade-off

For low-resource code-mixed text, label fidelity and scale are in tension. A corpus can be labelled carefully or it can be labelled in volume. The strategies reported by the studies reviewed here are best read not as competing solutions to one problem, but as different decisions about which of the two to sacrifice. On that reading, the strategies fall into three tiers: manual annotation, semi-automated annotation, and fully automated generative annotation.

The first tier is manual annotation by people. Two of the four Malay-English sentiment resources were built this way, and they are the only two whose labels come from human judgement. Kasmuri and Basiron (2019) drew 5,849 sentences from 23 personal blogs, selected 4,191 for annotation, and retained 2,316 in the finished corpus, reporting $\kappa = 0.83$ for the judgement of whether a sentence was opinionated or factual. Kong et al. (2023) collected 108,246 tweets and manually annotated 11,568 of them into three sentiment classes, using three independent annotators working from a written guideline. What both illustrate is the cost of the decision rather than a defect in the method. Kasmuri and Basiron kept 40% of the sentences they started with, and Kong et al. labelled 11% of what they collected. The labour that produces reliable labels is the same labour that prevents either corpus from reaching the scale a general-purpose Manglish resource would require.

The second tier is semi-automated annotation, which trades in the opposite direction by using existing tools to label at volume and accepting the tools' limitations as the price. Mohd Shamsuddin et al. (2024) built MESocSentiment this way. They collected 229,566 tweets carrying the "#Malaysia" keyword, reduced the set to 138,646 in pre-processing, applied the Malaya and Lingua libraries to identify code-switched posts, and arrived at 20,226 tweets. Sentiment labels were then assigned by majority vote among three existing analysers. The authors state the motivation plainly. The language identification libraries "reduced the need to find fluent speakers in Malay and English to classify the tweets". The reasoning is

defensible, and there is evidence from other labelling tasks that automated annotators can match or exceed human crowd workers (Gilardi et al., 2023). What the tier concedes is that the labels can be no more competent than the tools producing them, and inherit whatever those tools cannot read.

The third tier replaces those tools with generative models used directly as annotators. Because a generative model is not restricted to a fixed vocabulary, it can in principle interpret an unfamiliar spelling or an unseen slang term from context, which is the capability the second tier lacks. In principle this dissolves the trade-off. The peer-reviewed evidence does not yet support that reading. Jadhav et al. (2025) evaluated generative annotators on Marathi, an Indo-Aryan language of western India, across four tasks: sentiment analysis, two-class hate speech detection, four-class hate speech detection, and news classification. They found two generative large language models, GPT-4o and Llama 3.1 405B, trailing a fine-tuned BERT baseline by average accuracy margins of 10.2% and 14.1% respectively. BERT is a bidirectional transformer encoder pre-trained on unlabelled text. They conclude that fine-tuned transformers remain necessary where linguistic resources are scarce. Marathi is morphologically rich where Malay is not, so the result bounds the tier's promise rather than measuring it in this setting. That it is the only peer-reviewed study of the question available to this review is itself part of the finding. The third tier is the least evidenced of the three, and no study examined here evaluates generative annotation on Malaysian code-mixed text.

Table 1. Annotation Strategies for Malaysian Malay-English Code-Mixed Sentiment Data

Tier	Regional example	Scale achieved	Evidence of fidelity	What is conceded
Manual	Kasmuri and Basiron (2019)	2,316 sentences retained from 5,849 candidates, drawn from 23 blogs	$\kappa = 0.83$, for a binary opinionated-or-factual judgement	Scale
Manual	Kong et al. (2023)	11,568 tweets annotated from 108,246 collected	Three annotators and a written guideline; no coefficient reported	Scale
Semi-automated	Mohd Shamsuddin et al. (2024)	19,714 posts from 229,566 collected	None reported; the pipeline has no human annotation stage	Demonstrable fidelity
Fully automated, generative	None published for Malay-English	—	—	Reliability precisely where linguistic resources are scarce (Jadhav et al., 2025)

3.2 Two sources of neutral-label skew

The clearest regional evidence of what the second tier costs is the class distribution of the corpus it produced. MESocSentiment is labelled 78.23% neutral, 16.33% positive and 5.44% negative across 19,714 posts (Mohd Shamsuddin et al., 2024). Two reasons support reading that distribution as an artefact of the labelling method rather than a property of Malaysian discourse. The corpus was collected under the "#Malaysia" keyword across sports, politics and the arts, which are not topics on which social media users are typically neutral. A comparable corpus of Malaysian tweets labelled by human annotators is only 23.6% neutral (Kong et al., 2023). This section separates the two distinct causes at work in that skew, because they produce identical labels and require opposite responses.

The first cause is the behaviour of a scoring function that finds nothing to score. Lexicon-based sentiment tools work by matching each word or phrase in a text against a sentiment lexicon and aggregating the scores of the matched words into an overall score (Selvaraju et al., 2024). That score is then divided into classes by threshold. VADER, one of the three analysers used to build MESocSentiment, sets its classification thresholds at −0.05 and +0.05 on a normalised score running from −1 to 1 (Hutto and Gilbert, 2014). Two consequences follow from combining those two facts, and this review states them as an inference rather than as a reported finding. A text whose affective tokens are absent from the lexicon contributes little or nothing to the aggregate. Its score therefore remains within the neutral band, and it is returned as neutral. On that reading, neutral is not always a judgement that a text is affectively balanced. It can also be the default output when nothing was recognised.

The corpus authors' own error analysis indicates that the mechanism operates on this data. Alongside the corpus, Mohd Shamsuddin et al. (2024) present two small tables comparing the ensemble's label with the correct one. All three of the misclassified examples they give are neutral labels that should have been positive, and in all three the sentiment is carried by Malay: *memang cantik sangat*, *selamat deepavali* and *raih gelar juara*. In the correctly classified negative example, the sentiment is carried by an English word, *worthless*. Six author-selected examples are illustrative rather than statistical, and no error rate can be computed from them. Their uniformity is nonetheless difficult to dismiss: where the ensemble succeeded, the affect was in a language its tools covered, and in every case where it failed, the affect was not.

The second source has nothing to do with tooling. Some posts are genuinely hard to judge, and the annotators of these resources say so. Kasmuri and Basiron (2019) note that low agreement in sentiment annotation arises because sentiment texts are subjective, so evaluations differ from one annotator to another and are shaped by each annotator's background. Kong et al. (2023) make the same point about their own task, observing that tagging text with sentiment labels is highly subjective and influenced by personal beliefs, and that a clear guideline is needed for this reason. As Section 2.2 established, disagreement of this kind is not removed by collecting more judgements (Pavlick and Kwiatkowski, 2019).

Separating the two sources matters because the labels they produce are indistinguishable and the responses they require are opposite. A genuinely neutral post is correctly labelled and belongs in the corpus. A falsely neutral post is an error

that a better-resourced annotator would not have made. A single-model pipeline is structurally unable to tell them apart, because the evidence that would distinguish them, namely the affect the annotator could not read, is exactly the evidence that annotator does not have.

The consequence is more damaging than the aggregate percentage suggests, because a false neutral does not only inflate one class. It also depletes another. Every strongly polarised post absorbed into the neutral class is a training example removed from a minority class that a downstream classifier needs most. At a 5.44% negative share, the smallest class in the corpus is also the one most exposed to that erosion.

3.3 What consensus does with disagreement

Pooling several annotators is the response most often proposed when a single annotator is unreliable (Artstein and Poesio, 2008; Uma et al., 2021). MESocSentiment is evidence against that response, because it already was a consensus pipeline. Three sentiment analysers labelled every post and the majority label was assigned. The corpus skewed anyway.

The reason is visible in the corpus authors' own reporting, and it is not the one that might be expected. Voting cancels error only to the extent that voters fail independently, and the usual guarantee of independence is architectural difference. Table 2 shows that architectural difference was present and did not help.

Table 2. The Three Annotators Used to Build MESocSentiment, and the Labels Each Assigned (Mohd Shamsuddin et al., 2024)

Analyser	How the corpus authors describe it	Languages it is described as covering	Neutral labels assigned, of 19,714
TextBlob	"a Python library used to process textual data"	Not stated	14,305 (72.6%)
VADER	"a rule-based model … made to focus on social media texts"	Not stated	13,115 (66.5%)
pysentimiento	A toolkit "trained for sentiment analysis in English, Spanish, Portugeuse, and Italian"	English, Spanish, Portuguese, Italian	16,916 (85.8%)
Majority vote	—	—	15,422 (78.2%)

Two things follow. The first is that every analyser skewed neutral before any vote was taken, so the skew was not produced by the voting rule. The second is that the analyser skewing hardest, at 85.8%, is pysentimiento, which is the transformer-based tool of the three, built on the neural architecture that underlies most current language models. The shared weakness is therefore not shallow processing, since the deepest model performed worst. None of the three is described by the corpus authors as covering Malay, and one is explicitly described as trained on four other languages. The property the three voters share is the language they cannot read, and

a Malay affective token invisible to one is invisible to all. The ensemble does not cancel that error. It ratifies it, and unanimity produced this way is indistinguishable from confidence.

The second weakness concerns the posts the ensemble cannot agree on. Where all three analysers returned different labels there was no majority, and the authors state their response directly: ambiguous cases "were removed to ensure the corpus only contains tweets from absolute categories". This affected 512 posts, or 2.53% of the 20,226 labelled. The difficulty is not the proportion but the composition. A post on which three analysers disagree is one where the signal is contested, or is carried by material none of them reads confidently. That description fits the heavily code-switched and particle-laden cases Section 2 identified as the hardest to annotate. The rule therefore removes the examples a robust classifier would most need, and removes them systematically rather than at random.

The effect on the corpus can be computed from the paper's own tables, and it runs in the wrong direction. Removing the contested posts left the neutral count unchanged at 15,422 while reducing the total from 20,226 to 19,714, so the neutral share rose by roughly two percentage points. Taken with the first weakness, static voting has a double effect: it keeps the labels it is confident about and wrong about, and deletes the ones it is uncertain about and would have learned from.

The methodological literature on annotation supports treating that deletion as a loss rather than a cleanup. Uma et al. (2021) question the assumption that a single correct interpretation exists for every item. On that view, the 512 discarded posts were not noise removed from the corpus but signal removed from it.

What the reviewed literature does not contain is any evaluation of a better procedure on this data. No study examined here adjudicates annotator disagreement on Malaysian code-mixed sentiment text, and the premise that annotators of different types fail independently, which is what licenses aggregation in the first place, is asserted rather than measured on this material. The evidence assembled above suggests that independence fails when the annotators share a language gap, but a suggestion is not a measurement and this review does not treat it as one. Both points are recorded here and returned to in Section 6 as questions for future research.

4. Models as Annotation Instruments

Section 3 established that the choice of annotator determines which errors a corpus will contain. This section asks where each family of model is limited when used as an annotator. Two families are examined: fine-tuned multilingual encoders, and generative models used directly as annotators. The question is not which architecture classifies best, which belongs to the comparative benchmarking placed outside the scope in Section 1. It is what each family cannot read, because that limitation is inherited by every corpus it labels.

Fine-tuned multilingual encoders acquire their competence from exposure. An encoder here is a model that reads text and produces representations of it, rather than generating new text. Manukonda and Kodali (2025) illustrate the mechanism. They fine-tuned XLM-RoBERTa, a multilingual encoder model, over original, fully transliterated and partially transliterated versions of the same text, and report a

macro F1 of 0.90 on the shared-task test set. The training method was masked language modelling, in which the model predicts words hidden from the input. Macro F1 is the mean of the per-class F1 scores, and F1 is the harmonic mean of precision and recall. The model handled the variant spellings because it had been shown them in training. A corollary follows from the method rather than from any claim the authors make. An encoder resolves the variation it has been trained on, so the coverage of a fine-tuned annotator is bounded by the coverage of its fine-tuning data. Extending that coverage requires more annotated or transliterated regional data, which is the resource whose scarcity motivated automated annotation in the first place. Two limits should be noted: the study is a shared-task system description rather than a full investigation, and the task is fake news detection in Malayalam rather than sentiment analysis in Malay. What transfers is the mechanism, not the result.

Generative models locate their competence differently. Section 3.1 described the capability that follows from not being restricted to a fixed vocabulary. It is the capability the encoder lacks, and it is why generative annotation is attractive for a variety that changes faster than its resources. The competence is distributed unevenly across languages, and not in the way resource volume alone would predict. Jadhav et al. (2025) found frontier models trailing a fine-tuned BERT baseline by 10.2% and 14.1% accuracy when annotating Marathi. Adilazuarda et al. (2022) tested Indonesian code-mixed with four embedded languages, namely English, Sundanese, Javanese and Malay, and found robustness to English mixing lower than robustness to Javanese and Malay mixing across all models tested. They attribute the difference to the composition of the pre-training corpus. The direction of that result should be stated explicitly. The embedded language best represented in pre-training was the one the models handled least robustly. Robustness to code-mixing therefore does not follow directly from how much of a language a model has seen.

The operative point is not the relative standing of the two families but the shape of their respective limitations. The encoder is reliable on what it has been shown and blind to what it has not. The generative model reasons about the unfamiliar, but its robustness on a given language pair cannot be predicted from the volume of data behind either language. Selecting an annotator is therefore not selecting a level of quality. It is selecting a failure mode, and the labels a corpus contains carry the signature of whichever mode was selected.

5. Evaluating Annotation Reliability

Sections 3 and 4 concerned how labels are produced. This section concerns how the labels that result are shown to be sound. Reliability is normally demonstrated with a chance-corrected agreement coefficient, of which Cohen's κ is the most widely used (Cohen, 1960; Artstein and Poesio, 2008). The coefficient measures how far the agreement two annotators actually reach exceeds the agreement expected by chance. Magnitudes are then read against conventions. In computational linguistics, values above 0.8 are generally treated as good reliability, and values between 0.67 and 0.8 as allowing tentative conclusions (Artstein and Poesio, 2008).

Those conventions assume that expected chance agreement is stable across datasets, and it is not. Feinstein and Cicchetti (1990) show that an imbalance in the marginal totals of an agreement table can convert a high observed agreement into a low coefficient. Their worked example illustrates the effect. Holding observed agreement at 0.85, an expected chance agreement of 0.50 yields $\kappa = 0.70$, while an expected chance agreement of 0.78 yields $\kappa = 0.32$. The annotators agreed on the same 85% of items in both cases. Byrt et al. (1993) formalise the effect, showing that κ reflects the distribution of data across categories as well as agreement, and propose a prevalence-adjusted coefficient in response. Di Eugenio and Glass (2004) name the effect the prevalence problem, and report a case in which two coders agreeing on 90 of 100 items yield κ values ranging from -0.048 to 0.80, depending only on how the two categories are distributed. Artstein and Poesio (2008) state the consequence for annotation directly: when one category is very common, high agreement can result from indiscriminate coding rather than from an ability to distinguish between categories.

That is the condition Malaysian code-mixed sentiment data is in. Section 3 established that the labelling methods available for this text produce heavily skewed class distributions, the largest corpus standing at 78.23% neutral. A skew of that size raises expected chance agreement, which is the quantity the correction depends on. The property that makes the labels unrepresentative is therefore the property that makes the statistic used to check them difficult to read. The difficulty runs in both directions. A low coefficient may indicate unreliable annotation, or a dominant class. A high one may indicate reliable annotation, or indiscriminate coding. Neither reading can be ruled out from the coefficient alone. This is the sense in which the annotation problem and the evaluation problem are one problem rather than two, and the reason correcting either alone leaves the other in place.

Table 3 sets out what the studies introducing the four Malay-English sentiment resources report about the reliability of their annotation.

Table 3. Reliability Evidence Reported by the Four Malay-English Sentiment Corpora

Corpus	Annotators	Written guideline	Coefficient reported	What the coefficient measured
MESocSentiment (Mohd Shamsuddin et al., 2024)	None; three automatic analysers	Not applicable	None	—
MyCovid-Senti (Kong et al., 2023)	Three, independent	Yes	None	—
MY-EN-CS (Kasmuri and Basiron, 2019)	Two undergraduates proficient in Malay and English	Yes; the guideline is the paper's contribution	$\kappa = 0.83$	Whether a sentence is opinionated or factual

SentiBahasaRojak (Romadhona et al., 2022)	Five native Bahasa Rojak experts	Not stated	"Kappa" = 0.77; the variant is not specified	Whether a stock-market post is positive or negative; reported for the stock review subset only

Three observations follow. The studies introducing two of the four resources report no agreement coefficient at all, including one whose corpus was annotated by three people working from a written guideline. The two coefficients that are reported were obtained for binary judgements, and neither is the three-class polarity task for which the resources are used. Romadhona et al. (2022) report a kappa value without stating which coefficient it is. Their annotation used five experts with the final label set by majority vote, and Cohen's κ is defined for two annotators.

None of the four corpora examined here therefore supplies what a reliability claim requires: a coefficient of a stated type, computed for the task the labels are used for, over annotators whose number and qualification are recorded. A corpus that did supply them would still face the interpretation problem above, because the class distributions these methods produce are the distributions under which the coefficient becomes hard to read. Establishing reliability for this data is not only unreported. With the instruments currently in use, it is not clearly achievable.

6. Synthesis: How the Challenges Compound

Sections 2 to 5 examined linguistic structure, annotation practice, model capability and reliability measurement in turn, which is how the studies reviewed here are organised. Read separately they describe four difficulties. Read together they describe one sequence, in which each difficulty creates the conditions for the next and the sequence returns to its own starting point. Figure 1 sets it out. Establishing that this structure exists is the contribution this review claims, and the directions in the second half of this section are traced to points within it.

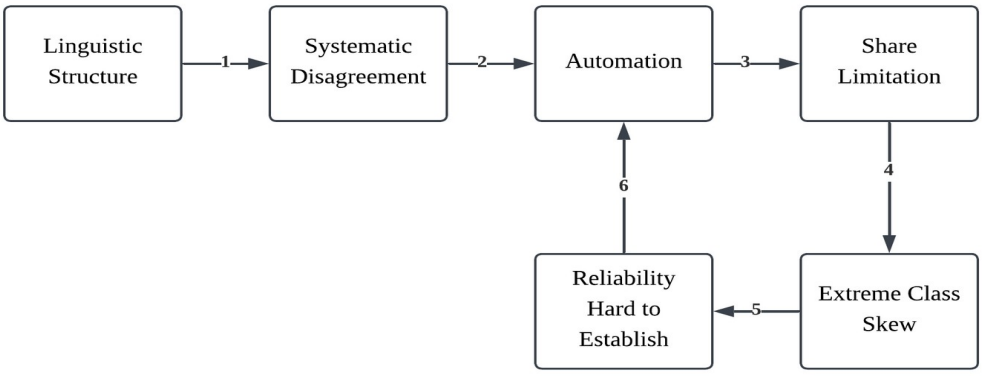


Figure 1. The Compounding Chain in Malaysian Code-Mixed Sentiment Annotation

The first link runs from linguistic structure to systematic disagreement. Section 2 established three properties of this text. The consequence is not only that agreement

is harder to reach. It is that disagreement takes a particular form. Where one annotator reads *lah* as a softener and another reads it as a marker of sarcasm, and each reading is stable, the disagreement recurs on the same construction in the same direction. Random disagreement is reduced by recruiting more annotators, because independent errors average out. Systematic disagreement is not, because additional annotators bring their own stable readings rather than noise around a shared one (Pavlick and Kwiatkowski, 2019). The usual response to unreliable annotation is therefore unsuited to the type of unreliability this text produces.

The second link runs from systematic disagreement to automation. Competent annotation of this material requires a fluent bilingual reader, and that requirement follows from the linguistic properties of the text rather than from any general fact about annotation. The corpus papers give cost and annotator availability as their reason for automating, and Mohd Shamsuddin et al. (2024) state that their tools "eliminated the need to find speakers fluent in Malay and English to label the dataset". Automation on this material is therefore downstream of the linguistic difficulty rather than an independent methodological preference.

The third link runs from automation to a shared limitation. The tools available for automation are those built for well-resourced languages. Section 3.3 documented the one regional case: none of the three analysers covers Malay, each skewed neutral before any vote was taken, and voting cancels error only to the extent that voters fail independently. Where the property they share is the language none of them reads, the ensemble ratifies the error instead of cancelling it.

The fourth link runs from the shared limitation to extreme class skew. False neutrals accumulate rather than being corrected, and the corpus acquires a class distribution that no participant intended. MESocSentiment reaches 78.23% neutral. Posts on which the three analysers could not agree were removed, 512 of them, on the stated ground of retaining only "absolute categories". Those posts are not a random sample. They are the contested, heavily code-switched cases that the first link identifies as the source of the difficulty, and removing them raised both the neutral share and the measured agreement.

The fifth link runs from extreme class skew to the difficulty of establishing reliability, and it is where the two halves of this review meet. The distribution produced at the fourth link is the specific condition under which a chance- corrected coefficient stops returning interpretable values (Feinstein and Cicchetti, 1990; Byrt et al., 1993). A researcher who suspects the labels are unsound and computes a coefficient obtains a low value, which is consistent with poor annotation and equally consistent with competent annotation of a skewed corpus. Raw agreement carries the same ambiguity in the opposite direction (Artstein and Poesio, 2008). The defect and the inability to detect the defect have one cause. That is what separates an unresolved problem from a merely difficult one: here the quality check is least informative exactly where quality is lowest.

The sixth link closes the cycle. Corpora whose reliability was never established are released, trained on and evaluated against, so every downstream comparison inherits an error term that nobody has quantified. The more consequential edge is the one that returns. The annotators used to build the next corpus are fine-tuned on

the last, and Section 4 showed that a fine-tuned encoder resolves only the variation it has been shown. The limitations of one generation of labels are transmitted to the instruments that produce the next, so the chain does not merely persist. It reproduces itself.

Read as a whole, the chain explains a pattern that the individual studies cannot. Each reviewed work intervenes at one link and does so competently. None addresses the chain as a chain, and an intervention at one link is assessed with an instrument compromised at another, so the assessment cannot establish whether the intervention worked. That is why effort in this area has been absorbed without accumulating.

The chain identifies where intervention would matter, not which intervention would succeed. Each of the five directions below is traced to a link, since a direction that cannot be traced to a finding is a preference rather than a conclusion.

The first direction is agreement measurement under class skew, and it follows from the fifth link. Prevalence-adjusted coefficients exist (Byrt et al., 1993) but have not been evaluated on code-mixed sentiment data, so it is not known which behaves best under the distributions these corpora actually exhibit. The study requires no new data collection, only an existing annotated corpus, a set of candidate coefficients, and a comparison of how they behave as skew increases. It is a precondition for the others, because until reliability can be measured under prevailing conditions, no corpus built hereafter can be shown to improve on the ones described here.

The second direction is a minimum reporting standard, and it follows from Table 3. A minimum set for any newly released corpus would comprise five elements: (1) the number of annotators and their stated qualification; (2) the availability of the annotation guideline; (3) the variant of the coefficient and the task it was computed for; (4) per-class rather than aggregate agreement; and (5) the full class distribution. Per-class agreement deserves particular emphasis, since an aggregate figure on a skewed corpus is dominated by the majority class and can conceal near-total disagreement on the minority polarities that matter most.

The third direction is a shared annotation convention for affect-bearing particles, and it follows from the first link. Because the disagreement these particles generate is systematic rather than random, the appropriate remedy is a shared convention rather than additional annotators, and no such convention exists for *lah*, *kan* and their equivalents. A model is available. Mohammad (2016) proposes a questionnaire that separates the speaker's emotional state, the target of the opinion and the attitude taken toward it. It also treats sarcasm, ridicule and mockery as a distinct response rather than folding them into polarity, which is precisely the ambiguity Section 2.2 identified. Kong et al. (2023) adapt this scheme into a four-question guideline for Malaysian tweets. What is absent is not the design but its adaptation to Malaysian particles and its adoption across projects.

The fourth direction is to publish label distributions rather than only resolved labels, and it follows from the fourth link. Current practice resolves disagreement by majority rule or removes it, discarding the information that the case was contested. Uma et al. (2021) report that models trained on the full distribution of annotator

judgements can outperform models trained on aggregated labels. Publishing that distribution would preserve the cases that current pipelines delete. It would also make the fourth link visible, since a corpus that reports what it discarded allows a reader to judge whether its reported reliability was earned or obtained by attrition.

The fifth direction is to test the independence premise, and it follows from the third link. Every consensus mechanism in this literature depends on the premise that annotators of different architecture fail independently, and on this material the premise is asserted throughout and measured nowhere. The study is a direct one: label a common sample with annotators of deliberately different competence, and measure the correlation of their errors rather than their individual accuracies. Low correlation would justify the aggregation strategies already in use. High correlation would indicate that consensus on this material adds confidence without adding evidence.

These are directions rather than results. The claim of this review is not that they will succeed, but that Figure 1 identifies where effort in this area is currently absorbed without accumulating.

7. Conclusion

This review examined one question: why reliable annotation and evaluation of Malaysian Malay-English code- mixed sentiment data remain unresolved. The answer usually given is architectural, and the evidence assembled here indicates that it is incomplete. Model choice accounts for part of the difficulty. The rest originates earlier, in the sentiment labels themselves and in the evidence offered that those labels are sound.

The contribution this review claims is not any of those findings taken alone but the relation between them. The linguistic properties of the text produce disagreement that is systematic rather than random, which makes fluent bilingual annotation necessary and expensive, which is why labelling is automated. The tools available for that automation cover neither Malay nor Manglish, so pooling them ratifies a shared error instead of cancelling it. The class skew that results is the precise condition under which the standard agreement coefficient stops returning interpretable values, so the defect and the inability to detect it have one cause rather than two. Each corpus built this way then trains the annotators that build the next. An intervention at any single point is assessed by an instrument compromised at another, which is why competent work at each link has not accumulated into progress.

The directions set out in Section 6 follow from that structure rather than from preference. The first of them, an agreement measure that behaves interpretably under the class distributions these corpora actually exhibit, is a precondition for evaluating the rest. Until reliability can be measured under the conditions this data presents, no corpus built hereafter can be shown to improve on the ones examined here.

References

Abu Bakar, M. F. R., Idris, N., Shuib, L., and Khamis, N. (2020). Sentiment analysis of noisy Malay text: State of art, challenges and future work. IEEE Access, 8, 24687–24696. https://doi.org/10.1109/ACCESS.2020.2968955

Adilazuarda, M. F., Cahyawijaya, S., Winata, G. I., Fung, P., and Purwarianti, A. (2022). IndoRobusta: Towards robustness against diverse code-mixed Indonesian local languages. Proceedings of the First Workshop on Scaling Up Multilingual Evaluation (SUMEval), 25–34.

AlAfnan, M. A., MohdZuki, S. F., and AlAfnan, S. M. (2025). Blending identities: Exploring the linguistic, cultural, and pragmatic dimensions of Manglish in the Malaysian society. Studies in Media and Communication, 13(3). https://doi.org/10.11114/smc.v13i3.7654

Artstein, R., and Poesio, M. (2008). Survey article: Inter-coder agreement for computational linguistics. Computational Linguistics, 34(4), 555–596. https://doi.org/10.1162/coli.07-034-R2

Bowman, S. R., and Dahl, G. (2021). What will it take to fix benchmarking in natural language understanding? Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 4843–4855. https://doi.org/10.18653/v1/2021.naacl-main.385

Byrt, T., Bishop, J., and Carlin, J. B. (1993). Bias, prevalence and kappa. Journal of Clinical Epidemiology, 46(5), 423–429. https://doi.org/10.1016/0895-4356(93)90018-V

Cohen, J. (1960). A coefficient of agreement for nominal scales. Educational and Psychological Measurement, 20(1), 37–46. https://doi.org/10.1177/001316446002000104

Coluzzi, P. (2017). Language planning for Malay in Malaysia: A case of failure or success? International Journal of the Sociology of Language, 2017(244), 17–38. https://doi.org/10.1515/ijsl-2016-0055

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., and Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 8440–8451. https://doi.org/10.18653/v1/2020.acl-main.747

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), 4171–4186. https://doi.org/10.18653/v1/N19-1423

Dewan Bahasa dan Pustaka. (2005). *Kamus Dewan* (4th ed.). Kuala Lumpur: Dewan Bahasa dan Pustaka.

Di Eugenio, B., and Glass, M. (2004). The kappa statistic: A second look. Computational Linguistics, 30(1), 95–101. https://doi.org/10.1162/089120104773633402

Doğruöz, A. S., Sitaram, S., Bullock, B. E., and Toribio, A. J. (2021). A survey of code-switching: Linguistic and social perspectives for language technologies. Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), 1654–1666. https://doi.org/10.18653/v1/2021.acl-long.131

Feinstein, A. R., and Cicchetti, D. V. (1990). High agreement but low kappa: I. The problems of two paradoxes. Journal of Clinical Epidemiology, 43(6), 543–549. https://doi.org/10.1016/0895-4356(90)90158-L

Frénay, B., and Verleysen, M. (2014). Classification in the presence of label noise: A survey. IEEE Transactions on Neural Networks and Learning Systems, 25(5), 845–869. https://doi.org/10.1109/TNNLS.2013.2292894

Gilardi, F., Alizadeh, M., and Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. Proceedings of the National Academy of Sciences, 120(30), e2305016120. https://doi.org/10.1073/pnas.2305016120

Hutto, C. J., and Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. Proceedings of the International AAAI Conference on Web and Social Media, 8(1), 216–225.

Jadhav, S., Shanbhag, A., Thakurdesai, A., Sinare, R., and Joshi, R. (2025). On limitations of LLM as annotator for low resource languages. Proceedings of the 8th International Conference on Natural Language and Speech Processing (ICNLSP-2025), 277–282.

Joshi, P., Santy, S., Budhiraja, A., Bali, K., and Choudhury, M. (2020). The state and fate of linguistic diversity and inclusion in the NLP world. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 6282–6293. https://doi.org/10.18653/v1/2020.acl-main.560

Kasmuri, E., and Basiron, H. (2019). Building a Malay-English code-switching subjectivity corpus for sentiment analysis. International Journal of Advanced Soft Computing and Its Applications, 11(1), 112–130.

Khanuja, S., Dandapat, S., Srinivasan, A., Sitaram, S., and Choudhury, M. (2020). GLUECoS: An evaluation benchmark for code-switched NLP. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 3575–3585. https://doi.org/10.18653/v1/2020.acl-main.329

Kong, J. T. H., Juwono, F. H., Ngu, I. Y., Nugraha, I. G. D., Maraden, Y., and Wong, W. K. (2023). A mixed Malay-English language COVID-19 Twitter dataset: A sentiment analysis. Big Data and Cognitive Computing, 7(2), 61. https://doi.org/10.3390/bdcc7020061

Manukonda, D. P., and Kodali, R. G. (2025). byteSizedLLM@DravidianLangTech 2025: Fake news detection in Dravidian languages using transliteration-aware XLM-RoBERTa and transformer encoder-decoder. Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages, 176–181.

Maskat, R., and Rahman, N. A. (2020). Categorization of Malay social media text and normalization of spelling variations and vowel-less words. International Journal on Advanced Science, Engineering and Information Technology, 10(4), 1380–1386.

Mohammad, S. M. (2016). A practical guide to sentiment annotation: Challenges and solutions. Proceedings of the 7th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis (WASSA), 174–179.

Mohd Shamsuddin, A. M., Samson Juan, S., Chua, S., and Bramantoro, A. (2024). Semi-automatic sentiment identification for Malay-English code-switched data. Journal of Advanced Research Design, 123(1), 198–212. https://doi.org/10.37934/ard.123.1.198212

Muysken, P. (2013). Language contact outcomes as the result of bilingual optimization strategies. Bilingualism: Language and Cognition, 16(4), 709–730. https://doi.org/10.1017/S1366728912000727

Nazir, A., Rao, Y., Wu, L., and Sun, L. (2022). Issues and challenges of aspect-based sentiment analysis: A comprehensive survey. IEEE Transactions on Affective Computing, 13(2), 845–863. https://doi.org/10.1109/TAFFC.2020.2970399

Northcutt, C. G., Athalye, A., and Mueller, J. (2021). Pervasive label errors in test sets destabilize machine learning benchmarks. Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS): Datasets and Benchmarks Track.

Pavlick, E., and Kwiatkowski, T. (2019). Inherent disagreements in human textual inferences. Transactions of the Association for Computational Linguistics, 7, 677–694. https://doi.org/10.1162/tacl_a_00293

Pontiki, M., Galanis, D., Papageorgiou, H., Androutsopoulos, I., Manandhar, S., AL-Smadi, M., Al-Ayyoub, M., Zhao, Y., Qin, B., De Clercq, O., Hoste, V., Apidianaki, M., Tannier, X., Loukachevitch, N., Kotelnikov, E., Bel, N., Jiménez-Zafra, S. M., and Eryiğit, G. (2016). SemEval-2016 Task 5: Aspect based sentiment analysis. Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016), 19–30. https://doi.org/10.18653/v1/S16-1002

Poria, S., Hazarika, D., Majumder, N., and Mihalcea, R. (2023). Beneath the tip of the iceberg: Current challenges and new directions in sentiment analysis research. IEEE Transactions on Affective Computing, 14(1), 108–132. https://doi.org/10.1109/TAFFC.2020.3038167

Romadhona, N. P., Lu, S.-E., Lu, B.-H., and Tsai, R. T.-H. (2022). BRCC and SentiBahasaRojak: The first Bahasa Rojak corpus for pretraining and sentiment analysis dataset. Proceedings of the 29th International Conference on Computational Linguistics (COLING), 4418–4428.

Selvaraju, S., Sjarif, N. N. A., and Anuar, M. S. M. (2024). Sentiment analysis in code mixing Malay text: A review. 2024 5th International Conference on Artificial Intelligence and Data Sciences (AiDAS), 280–285. https://doi.org/10.1109/AiDAS63860.2024.10730298

Treffers-Daller, J., Majid, S., Thai, Y. N., and Flynn, N. (2022). Explaining the diversity in Malay-English code-switching patterns: The contribution of typological similarity and bilingual optimization strategies. Languages, 7(4), 299. https://doi.org/10.3390/languages7040299

Uma, A. N., Fornaciari, T., Hovy, D., Paun, S., Plank, B., and Poesio, M. (2021). Learning from disagreement: A survey. Journal of Artificial Intelligence Research, 72, 1385–1470. https://doi.org/10.1613/jair.1.12752

Winata, G. I., Aji, A. F., Yong, Z. X., and Solorio, T. (2023). The decades progress on code-switching research in NLP: A systematic survey on trends and challenges. Findings of the Association for Computational Linguistics: ACL 2023, 2936–2978. https://doi.org/10.18653/v1/2023.findings-acl.185

Winata, G. I., Cahyawijaya, S., Liu, Z., Lin, Z., Madotto, A., and Fung, P. (2021). Are multilingual models effective in code-switching? Proceedings of the Fifth Workshop on Computational Approaches to Linguistic Code-Switching, 142–153. https://doi.org/10.18653/v1/2021.calcs-1.20